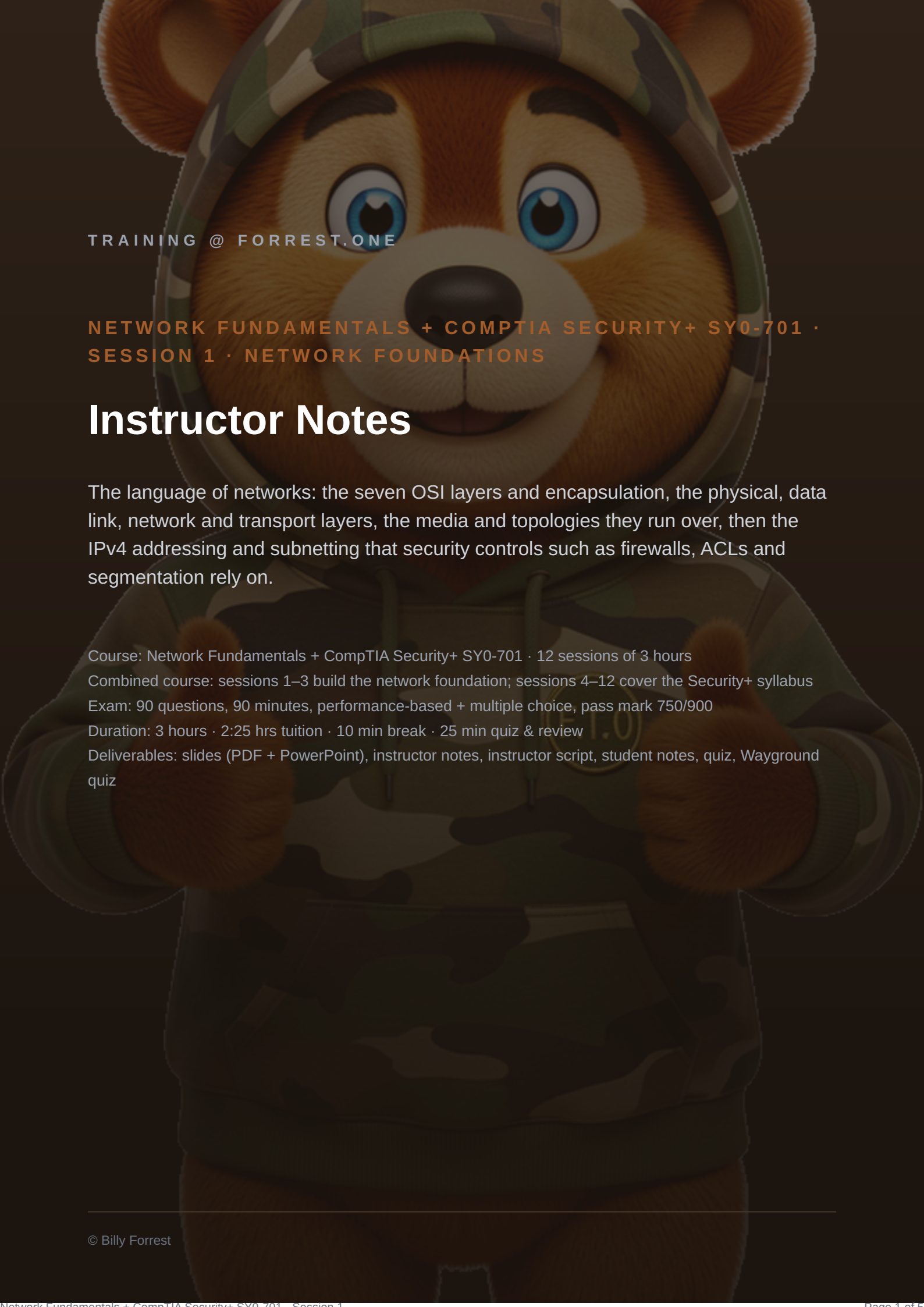




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·  
SESSION 1 · NETWORK FOUNDATIONS**

# Instructor Notes

The language of networks: the seven OSI layers and encapsulation, the physical, data link, network and transport layers, the media and topologies they run over, then the IPv4 addressing and subnetting that security controls such as firewalls, ACLs and segmentation rely on.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Combined course: sessions 1–3 build the network foundation; sessions 4–12 cover the Security+ syllabus

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, quiz, Wayground quiz

## Instructor Notes — Network Foundations: Networking Models & IP Addressing

A 3-hour delivery guide for Session 1. This session builds the network vocabulary the whole course depends on, and it sets the security tone from the start: every layer, every medium and every subnet is introduced alongside the control or the attack it relates to. Keep the whiteboard busy. Draw the OSI stack, walk a message down it, sketch the star and mesh, and work a subnet mask in front of the class before releasing them to practise.

### Session at a glance

| Time      | Block                            | Slides | Focus / activity                                                           |
|-----------|----------------------------------|--------|----------------------------------------------------------------------------|
| 0:00–0:15 | Welcome & course map             | 1–3    | Two-strand journey; objectives                                             |
| 0:15–0:35 | Chapter 1 & OSI overview         | 4–7    | Why layers exist; seven layers; a message's journey; encapsulation         |
| 0:35–1:00 | Chapter 3 · Physical & Data Link | 8–12   | Signals and media; MAC addressing; sublayers; the layer 1 first check      |
| 1:00–1:20 | Chapter 4 · Network & Transport  | 13–17  | TTL and routers; TCP against UDP; upper layers; discussion                 |
| 1:20–1:30 | Break                            | 18     | 10-minute break                                                            |
| 1:30–1:50 | Chapter 5 · Media & Topologies   | 19–24  | Wireless and wired media; connectors; star, mesh, hybrid                   |
| 1:50–2:20 | Chapter 6 · IPv4 & Subnetting    | 25–31  | Structure; reserved ranges; masks; classless prefixes; arithmetic activity |
| 2:20–2:50 | Quiz & review                    | 32     | Quiz_Session_01.html, 20 questions                                         |
| 2:50–3:00 | Close                            | 33–35  | Takeaways; checklist; Session 2 preview                                    |

### Materials and preparation

- Projector or screen with the Session 1 slide deck (02\_Slides.pdf or 02\_Slides.pptx).

- One computer or tablet per pair for Quiz\_Session\_01.html (any browser, no internet required).
- Printed or digital copy of the Student Notes for each candidate.
- Whiteboard or flipchart with the OSI stack and a subnetting grid drawn ready.
- Optional: a short length of twisted pair, an RJ45 plug and an SC or LC fibre connector to pass round during Chapter 5.

## Delivery guidance by slide

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### Slides 1–3 · Welcome (15 minutes)

Set the course shape early: three network-foundation sessions, then nine security sessions. Explain why networks come first: every firewall rule, IDS signature and VPN tunnel speaks network. Walk the ten objectives, noting that no prior networking knowledge is assumed, and point out that the quiz at the end tests every one of them.

### Slides 4–7 · Chapter 1 and the OSI overview (20 minutes)

Open with the troubleshooting argument on slide 4, because it earns the model's place before you teach any layer names. Then draw the seven layers as you name them, one job each. Slide 6 is the pivot: walk one message from the application down to the wire, saying what is added and which address is used at each step. Slide 7 then names that process as encapsulation and fixes the vocabulary: data, segment or datagram, packet, frame, bits. Exam questions use those words precisely, so make candidates say them.

### Slides 8–12 · Chapter 3, Physical and Data Link (25 minutes)

Layer 1 is short on theory and long on practice: three signal types, three media, and the standards for cabling, connectors and signalling. Slide 9 turns the three media into a trade-off table: cost, distance, interference. Slide 10 introduces Media Access Control: 48 bits, hexadecimal, the frame, and the Cyclic Redundancy Check. Slide 11 covers the two sublayers; the memory hook is that Logical Link Control looks up to the network layer and Media Access Control looks down to the physical. Slide 12 is a habit worth twenty minutes of anyone's career: a completely dead link is almost always physical, so check the cable, the port light and the link state before touching a routing table.

#### Security thread for Chapter 3

Media Access Control addresses are set by the manufacturer but changeable in software, so a hardware address is not an identity. This is the seed for ARP poisoning, MAC flooding and port security in later sessions. Say it here; examiners test it later.

### Slides 13–17 · Chapter 4, Network, Transport and Above (20 minutes)

Slide 13 draws the distinction that matters most: a hardware address identifies a device on a local segment, an IP address identifies a device across networks. Add Time to Live as the loop-stopper, and route adaptation as the reason the internet survives partial failures. Slide 14 introduces ports as application identifiers, then sets up the question to ask: reliability or speed? Slide 15 answers it in a table; take TCP and UDP one row at

a time rather than reading the table out. Slide 16 covers layers 5, 6 and 7 separately, and corrects the common error that the application layer is the program rather than the protocol layer.

### Discussion prompt (slide 17)

"Name the layer first, then the control presents itself. Why does that work?" A web attack is layer 7, so the answer is a web application firewall; a spoofed packet is layer 3, so filter at the router; a rogue switch or a poisoned address table is layer 2, so use port security. The reason it works is that each layer has one job, so each layer has its own attacks and its own defences.

### Slide 18 · Break (10 minutes)

Announce the second half: media and topologies, then IPv4 addressing and subnetting. Tell candidates to bring a calculator if they want one, though the arithmetic is all mental.

### Slides 19–24 · Chapter 5, Media and Topologies (20 minutes)

Wireless first (slide 19). The one idea to land is that wireless is a shared medium: lower frequencies travel further and carry less, higher frequencies carry more over less distance, and every device on the channel competes for the same airtime. Slide 20 covers twisted pair, categories, shielding, coaxial and fibre; close with the rule of thumb that a link between two buildings is almost always fibre. Slide 21 gives the decision aid: distance, bandwidth, interference. Slide 22 is the connector and transceiver slide; pass a physical RJ45 and a fibre connector round if you have them, and explain why hot-swappable Small Form factor Pluggable modules matter operationally. Slides 23 and 24 handle topology. Star is easy to troubleshoot but has a single point of failure; full mesh has no single point of failure but costs and complexifies; hybrid meshes what needs resilience and stars the rest. Add spine and leaf as the data centre pattern with predictable latency.

### Slides 25–31 · Chapter 6, IPv4 Addressing and Subnetting (30 minutes)

Start with structure, not masks: 32 bits, four octets, dotted decimal, each octet 0 to 255, about 4.29 billion addresses, which is why nobody expected exhaustion. Slide 26 is recognition practice: the three private ranges, loopback, and 169.254. Be explicit that a 169.254 address is a symptom of a failed address request, not a configuration, because candidates misread it constantly. Slide 27 introduces the mask as a run of ones then zeros, with two addresses always reserved. Work slide 28's example on the board rather than reading it: address 192.168.1.50 with mask 255.255.255.0 gives network 192.168.1.0, hosts .1 to .254, broadcast 192.168.1.255. Slide 29 covers classless addressing, including the counter-intuitive rule that a larger prefix means a smaller network, and introduces VLSM as per-subnet sizing, largest first. Slide 30 is the table to memorise: /24 to /30 with masks and usable hosts, noting that every extra prefix bit halves the count and two addresses are always lost.

### Activity (slide 31)

Five quick calculations, 6 minutes solo then 4 minutes debrief. Answers: 192.168.10.48; 6 usable hosts; /28; 4 subnets of 62 hosts each; the network or broadcast address. Circulate and correct technique, not just answers.

## Slides 32–35 · Quiz & close (25 minutes)

Run Quiz\_Session\_01.html (slide 32), review the low-scoring topic areas against the tags, walk the takeaways on slide 33, assign the revision checklist on slide 34, then preview Session 2: switching, VLANs and Spanning Tree Protocol.

## Common misconceptions and corrections

| Misconception                                | Correction                                                                                        |
|----------------------------------------------|---------------------------------------------------------------------------------------------------|
| "The OSI model is theoretical"               | It maps to real controls: firewalls read layer 3 and 4, web application firewalls work at layer 7 |
| "Layering is just an exam diagram"           | Each layer has one job, which is what makes faults testable one at a time                         |
| "The application layer is the application"   | It is the protocol layer, where HTTP and DNS live, not the program the user runs                  |
| "TCP and UDP are both just transport"        | TCP is connection oriented, ordered and retransmits; UDP is connectionless and lower latency      |
| "A MAC address proves identity"              | It is changeable in software, so it is not a trustworthy identity                                 |
| "Fibre is always the right choice"           | It is right for distance and interference; twisted pair wins on cost for short office runs        |
| "A mesh network has no downsides"            | Full mesh has no single point of failure but maximum cost and complexity                          |
| "A /24 always means 254 addresses"           | 254 is the usable host count only when the whole block is one subnet                              |
| "The network address can be assigned"        | The first address is the network identifier, the last is the broadcast, and neither is assignable |
| "A 169.254 address is a valid configuration" | It is self-assigned when no server answered, so treat it as a fault symptom                       |
| "A larger prefix means a bigger network"     | The opposite: /30 is two addresses, /24 is 254 usable hosts                                       |

## Assessment check

- Session objectives map to the six chapter headings and the course's network-foundation strand.
- The HTML quiz tags every question with its topic area for traceable revision.
- The Wayground quiz (15 questions) gives an in-class, importable assessment of the same topics.
- Ask each candidate to state one layer, one medium and one mask rule from memory before they leave, as an exit check.